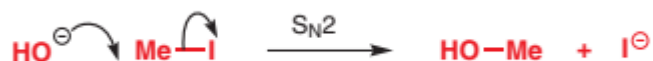


1. (1) Is the following reaction a concerted reaction or not? Why?



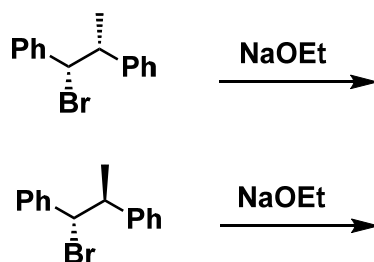
(2) If a compound with R configuration undergoes an SN2 reaction, does it always give a compound with S configuration as the product? Explain your answer.

2. Considering the following reactions:

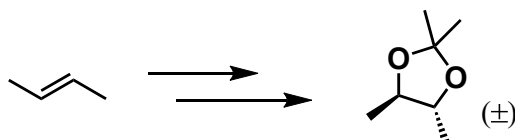


If the reaction is run at $-80\text{ }^{\circ}\text{C}$ for 1 h, the ratio of B to C is 4 to 1; if the reaction is run at $60\text{ }^{\circ}\text{C}$ for 1 h, the ratio of B to C is 3 to 1. It is known that for the reaction $B \leftrightarrow C$ at $60\text{ }^{\circ}\text{C}$, the equilibrium constant K equals 5. Out of B and C, which one is the thermodynamic product and which one is the kinetic product? Explain your answer and explain the product ratios at these two temperatures.

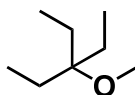
3. Give the major products for the following reaction and explain the stereochemistry:



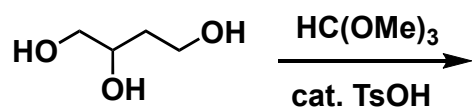
4. Using necessary inorganic or organic reagents, complete the following transformation.



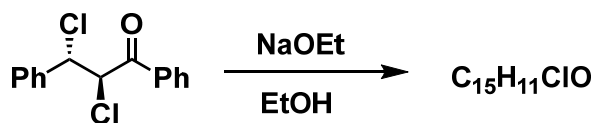
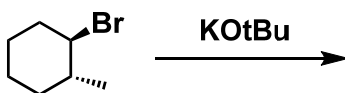
5. Using alkanes and alcohols containing less than 4 carbon atoms (3C and below) along with necessary inorganic reagents, suggest an efficient synthetic route for the following molecule.



6. Give the structure of the major product for the following reaction and draw the mechanism:



7. Give the structure of the major product for each of the following reactions and explain the stereochemistry:



8. Suggest a mechanism for the following reaction. The product has these signals in its ^1H NMR spectrum: 3.9 (1H, dqd, $J = 12, 7, 4$) and 4.3 (1H, dd $J = 11, 3$). Please draw the structure of this product and assign these signals.

